THE AUTOMORPHISM GROUP OF $X_1(42)$ OVER $\mathbb{F}_5$: EXPLICIT FORMULAS

SOURAJYOTI MAITI

This file accompanies the paper *The gonality of $X_1(42)$ is* 12 and records all twelve elements of $\mathrm{Aut}_{\mathbb{F}_5}(X)$, $X = X_1(42)_{\mathbb{F}_5}$, as computed by Magma's `Automorphisms`.

Conventions

Magma represents the function field of X (built from Sutherland's `FFFc42` model) as

$$\mathbb{F}_5(X) = \mathbb{F}_5(t)[y]/\big(F(t,y)\big), \qquad [\mathbb{F}_5(X) : \mathbb{F}_5(t)] = 16,$$

where t generates the rational subfield and y is the generator `AFF.1`. An automorphism σ is determined by $\sigma(t)$ and $\sigma(y)$, each written uniquely as $\sum_{j=0}^{15} c_j(t)\, y^j$ with $c_j \in \mathbb{F}_5(t)$.

After cancellation, every denominator that occurs is of the form $t^a(t+4)^b(t^2+4t+1)^c = t^a(t-1)^b(t^2-t+1)^c$. We write $c_j = N_j(t)/\big(t^{a_j}(t+4)^{b_j}(t^2+4t+1)^{c_j}\big)$ and record, for each j, the exponents a_j, b_j, c_j and the coefficients of $N_j \in \mathbb{F}_5[t]$ from the leading coefficient down to the constant term; a row with entry 0 means $c_j = 0$. For example, a row $j \mid 2\ 1\ 0 \mid 2 \mid 3\ 0\ 1$ stands for $c_j = (3t^2+1)/\big(t^2(t+4)\big)$.

The group is cyclic of order 12, generated by σ_4, and acts simply transitively on the twelve rational places $P_1, \dots, P_{12}$ (numbered as in Magma's `Places(AFF,1)`). Table 1 gives the order of each σ_i, its expression as a power of σ_4, whether it lies in the diamond subgroup C, and its permutation of the rational places; Table 2 lists the orbits of each cyclic subgroup $\langle\sigma_i\rangle$ on the rational places.

Table 1. The automorphisms σ_i: order, expression as a power of σ_4, membership in the diamond subgroup C, and the induced permutation of the rational places $P_1, \dots, P_{12}$ (numbered as in Magma's `Places(AFF,1)`), in cycle notation on the indices. Here $Q_1 = P_{12}$ and $Q_2 = P_1$.

σ	order	σ_4^k	in C	permutation of $X(\mathbb{F}_5)$
σ_1	1	σ_4^0	yes	id
σ_2	4	σ_4^9	no	(1 11 6 2)(3 4 7 12)(5 9 10 8)
σ_3	6	σ_4^{10}	yes	(1 10 7 6 5 3)(2 9 4 11 8 12)
σ_4	12	σ_4^1	no	(1 9 3 2 5 12 6 8 7 11 10 4)
σ_5	3	σ_4^8	yes	(1 7 5)(2 4 8)(3 10 6)(9 11 12)
σ_6	2	σ_4^6	yes	(1 6)(2 11)(3 7)(4 12)(5 10)(8 9)
σ_7	3	σ_4^4	yes	(1 5 7)(2 8 4)(3 6 10)(9 12 11)
σ_8	12	σ_4^5	no	(1 12 10 2 7 9 6 4 5 11 3 8)
σ_9	12	σ_4^{11}	no	(1 4 10 11 7 8 6 12 5 2 3 9)
σ_{10}	6	σ_4^2	yes	(1 3 5 6 7 10)(2 12 8 11 4 9)
σ_{11}	4	σ_4^3	no	(1 2 6 11)(3 12 7 4)(5 8 10 9)
σ_{12}	12	σ_4^7	no	(1 8 3 11 5 4 6 9 7 2 10 12)

The whole group $\mathrm{Aut}_{\mathbb{F}_5}(X) = \langle\sigma_4\rangle$ has a single orbit on $X(\mathbb{F}_5)$, and the diamond subgroup $C = \langle\sigma_3\rangle = \langle\sigma_{10}\rangle$ has the two orbits $\{P_2, P_4, P_8, P_9, P_{11}, P_{12}\} \ni Q_1$ and $\{P_1, P_3, P_5, P_6, P_7, P_{10}\} \ni Q_2$.

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TABLE 2. The orbits of the cyclic subgroup $\langle\sigma_i\rangle$ on the rational places $X(\mathbb{F}_5) = \{P_1, \ldots, P_{12}\}$. Since $\mathrm{Aut}_{\mathbb{F}_5}(X)$ acts freely on $X(\mathbb{F}_5)$, all orbits of $\langle\sigma_i\rangle$ have size equal to the order of σ_i. Here $Q_1 = P_{12}$ and $Q_2 = P_1$.

σ	#orbits	size	orbits of $\langle\sigma\rangle$ on $X(\mathbb{F}_5)$
σ_1	12	1	$\{P_1\}$, $\{P_2\}$, $\{P_3\}$, $\{P_4\}$, $\{P_5\}$, $\{P_6\}$, $\{P_7\}$, $\{P_8\}$, $\{P_9\}$, $\{P_{10}\}$, $\{P_{11}\}$, $\{P_{12}\}$
σ_2	3	4	$\{P_1, P_2, P_6, P_{11}\}$, $\{P_3, P_4, P_7, P_{12}\}$, $\{P_5, P_8, P_9, P_{10}\}$
σ_3	2	6	$\{P_1, P_3, P_5, P_6, P_7, P_{10}\}$, $\{P_2, P_4, P_8, P_9, P_{11}, P_{12}\}$
σ_4	1	12	$\{P_1, P_2, P_3, P_4, P_5, P_6, P_7, P_8, P_9, P_{10}, P_{11}, P_{12}\}$
σ_5	4	3	$\{P_1, P_5, P_7\}$, $\{P_2, P_4, P_8\}$, $\{P_3, P_6, P_{10}\}$, $\{P_9, P_{11}, P_{12}\}$
σ_6	6	2	$\{P_1, P_6\}$, $\{P_2, P_{11}\}$, $\{P_3, P_7\}$, $\{P_4, P_{12}\}$, $\{P_5, P_{10}\}$, $\{P_8, P_9\}$
σ_7	4	3	$\{P_1, P_5, P_7\}$, $\{P_2, P_4, P_8\}$, $\{P_3, P_6, P_{10}\}$, $\{P_9, P_{11}, P_{12}\}$
σ_8	1	12	$\{P_1, P_2, P_3, P_4, P_5, P_6, P_7, P_8, P_9, P_{10}, P_{11}, P_{12}\}$
σ_9	1	12	$\{P_1, P_2, P_3, P_4, P_5, P_6, P_7, P_8, P_9, P_{10}, P_{11}, P_{12}\}$
σ_{10}	2	6	$\{P_1, P_3, P_5, P_6, P_7, P_{10}\}$, $\{P_2, P_4, P_8, P_9, P_{11}, P_{12}\}$
σ_{11}	3	4	$\{P_1, P_2, P_6, P_{11}\}$, $\{P_3, P_4, P_7, P_{12}\}$, $\{P_5, P_8, P_9, P_{10}\}$
σ_{12}	1	12	$\{P_1, P_2, P_3, P_4, P_5, P_6, P_7, P_8, P_9, P_{10}, P_{11}, P_{12}\}$

The tables below are generated mechanically by the script `aut_to_latex.py` from the data file `X1_42_aut_F5.txt`. The script checks the factorization of every denominator exactly, and the tables are an exact transcription of Magma's output.

The twelve automorphisms

The automorphism σ_1.
Orbits of $\langle\sigma_1\rangle$ on $X(\mathbb{F}_5)$: $\{P_1\}$, $\{P_2\}$, $\{P_3\}$, $\{P_4\}$, $\{P_5\}$, $\{P_6\}$, $\{P_7\}$, $\{P_8\}$, $\{P_9\}$, $\{P_{10}\}$, $\{P_{11}\}$, $\{P_{12}\}$.

$\sigma_1(t) = t$ and $\sigma_1(y) = y$.

The automorphism σ_2.
Orbits of $\langle\sigma_2\rangle$ on $X(\mathbb{F}_5)$: $\{P_1, P_2, P_6, P_{11}\}$, $\{P_3, P_4, P_7, P_{12}\}$, $\{P_5, P_8, P_9, P_{10}\}$.

$$\sigma_2(t) = \sum_{j=0}^{15} \frac{N_j(t)}{t^{a_j}(t+4)^{b_j}(t^2+4t+1)^{c_j}}\, y^j$$

j	a_j	b_j	c_j	$\deg N_j$	coefficients of N_j, from $t^{\deg N_j}$ down to t^0
0	19	10	2	33	4442242234042043401421442232113321
1	20	11	2	34	42341204033012300413401434242121443
2	21	11	2	35	41314023401043242201112230303204324 2
3	22	10	2	33	21230012243330001122110211041311 23
4	23	12	2	36	12114434420202120204212434333443013 11
5	29	15	2	44	14001112421134423010311010030104401444441332 1
6	30	16	2	45	40304400002200040044212443402411331331111021443
7	31	16	2	44	13134121121130310203401241413404322140432400 2
8	32	16	2	43	32314042213024102233444334301401223040340221
9	28	16	2	37	22200431300112101431301042230044002241
10	29	16	2	38	43404121341011434343143414123031000121 3
11	30	16	2	38	41121122314043310012321443143132044103 2
12	31	16	2	37	41121111120300002100100311030034103303
13	32	16	2	37	22434132232031332442223424124100414334
14	28	16	2	31	33340434000331322323344121400232
15	29	16	2	31	44230410344443013130344201300114

$$\sigma_2(y) = \sum_{j=0}^{15} \frac{N_j(t)}{t^{a_j}(t+4)^{b_j}(t^2+4t+1)^{c_j}}\, y^j$$

j	a_j	b_j	c_j	$\deg N_j$	coefficients of N_j, from $t^{\deg N_j}$ down to t^0
0	18	10	3	35	1 3 3 2 1 2 3 4 0 0 4 0 2 0 3 4 0 2 3 4 0 4 2 0 4 4 3 1 2 3 3 4 0 0 3 4
1	19	11	3	36	1 4 1 1 3 2 3 1 1 3 4 1 1 0 3 0 1 3 2 3 3 0 1 2 4 4 3 4 1 2 4 3 4 2 0 1 2
2	20	11	3	35	3 1 3 3 2 4 1 1 3 4 4 4 2 3 2 4 1 2 4 1 3 4 1 3 2 2 4 0 4 0 1 4 3 4 1 3
3	21	10	3	35	4 4 0 1 4 0 0 2 1 0 3 3 0 3 3 2 3 3 0 3 2 1 2 3 1 1 2 2 3 4 3 0 1 3 3 2
4	22	12	3	35	1 0 0 1 4 0 1 3 3 3 4 2 2 2 4 1 4 2 0 4 3 2 2 4 2 3 3 4 1 4 3 3 4 0 4 4
5	28	15	3	45	1 3 0 4 2 4 4 2 0 3 3 0 1 2 3 3 3 4 4 3 0 2 4 4 0 1 4 3 0 1 1 1 4 3 0 4 4 1 0 1 0 4 0 0 3 4
6	29	16	3	46	1 2 3 4 1 0 0 0 3 0 1 4 4 0 1 3 0 1 4 2 1 3 0 0 0 0 4 2 3 2 3 3 3 4 4 1 2 2 2 2 0 4 4 2 0 1 2
7	30	16	3	45	4 2 4 4 4 4 0 1 4 2 2 0 1 3 1 1 0 2 3 2 4 0 0 2 0 2 4 4 3 1 3 4 3 0 4 2 4 3 4 0 4 2 0 1 0 3
8	31	16	3	44	4 0 0 2 3 4 3 2 4 3 0 1 3 3 3 4 0 1 3 1 3 0 0 1 3 0 2 3 2 0 1 4 2 1 1 1 2 2 1 0 3 3 1 3 4
9	27	16	3	38	2 3 2 3 2 4 3 3 3 4 2 1 4 2 3 4 0 1 3 0 1 0 1 0 0 1 4 4 2 1 3 4 4 3 1 2 1 1 4
10	28	16	3	39	3 1 1 4 4 3 3 2 2 2 1 2 3 0 1 4 2 4 4 1 3 0 0 0 4 0 1 3 4 2 0 1 3 1 4 4 3 2 4 2
11	29	16	3	39	1 4 3 1 3 0 3 0 4 4 1 2 2 2 3 2 2 1 3 4 4 1 2 1 2 4 2 1 4 1 4 0 3 4 0 1 2 1 2 3
12	30	16	3	38	2 0 4 0 1 2 3 0 0 0 3 1 2 1 0 4 3 3 2 2 1 3 1 4 0 4 0 3 0 0 4 3 3 4 0 3 1 0 2
13	31	16	3	38	3 2 2 3 0 2 3 2 1 4 3 0 1 2 0 1 0 2 2 2 0 4 0 1 1 1 1 2 2 3 1 3 2 0 2 3 4 2 1
14	27	16	3	32	4 4 1 2 1 3 1 1 4 2 2 0 1 2 0 1 4 2 2 3 4 0 2 1 2 3 3 1 1 3 4 2 3
15	28	16	3	32	1 0 0 4 1 3 4 4 3 0 2 4 0 2 1 1 3 3 4 0 2 4 3 0 1 0 1 1 3 1 1 4 1

The automorphism σ_3.

Orbits of $\langle\sigma_3\rangle$ on $X(\mathbb{F}_5)$: $\{P_1, P_3, P_5, P_6, P_7, P_{10}\}$, $\{P_2, P_4, P_8, P_9, P_{11}, P_{12}\}$.

$$\sigma_3(t) = \sum_{j=0}^{15} \frac{N_j(t)}{t^{a_j}(t+4)^{b_j}(t^2+4t+1)^{c_j}}\, y^j$$

j	a_j	b_j	c_j	$\deg N_j$	coefficients of N_j, from $t^{\deg N_j}$ down to t^0
0	12	13	0	24	1 4 1 4 0 4 2 2 2 4 3 3 2 0 2 4 3 1 3 4 1 4 1 2 4
1	13	14	0	25	1 4 1 0 2 3 0 0 4 4 1 1 0 0 1 2 0 1 1 1 0 1 3 0 3 2
2	14	14	0	25	1 2 4 2 3 3 2 3 4 3 3 2 4 0 4 1 0 2 0 4 3 4 0 1 4 3
3	15	13	0	24	1 3 0 0 2 1 0 4 3 3 0 4 0 4 3 4 3 1 3 3 1 4 0 0 2
4	16	15	0	27	4 2 2 1 4 3 1 2 0 0 1 3 2 4 1 2 3 4 0 0 0 0 0 1 0 2 3 4
5	22	18	0	34	3 4 2 3 1 0 4 3 4 1 0 1 2 2 1 2 0 2 1 4 4 4 4 1 0 0 1 1 3 1 1 4 1 2 4
6	23	19	0	35	3 0 2 1 2 4 1 3 0 4 0 1 3 0 1 1 2 0 3 4 4 3 4 4 1 0 2 1 4 3 1 1 3 0 3 2
7	24	19	0	34	4 2 4 3 1 4 3 4 0 1 4 1 0 4 0 3 2 0 0 0 1 0 3 1 0 2 2 4 0 3 1 2 2 3 3
8	25	19	0	34	3 1 1 1 3 4 1 1 2 2 2 3 4 1 2 1 3 2 0 3 1 2 0 2 4 1 1 3 1 3 0 3 2 2 4
9	21	19	0	29	4 2 0 1 0 3 2 3 2 2 2 1 0 1 2 4 0 1 4 4 0 0 4 1 4 3 3 0 0 4
10	22	19	0	28	4 1 2 3 2 4 1 0 2 3 0 1 0 4 0 4 2 4 0 3 3 3 4 4 2 2 0 1 2
11	23	19	0	27	1 1 3 4 3 1 1 3 2 1 4 0 0 1 1 0 1 3 3 1 2 2 0 1 3 4 0 3
12	24	19	0	26	2 1 2 3 2 4 0 1 1 3 1 2 1 3 0 0 3 1 4 3 4 2 3 3 0 2 2
13	25	19	0	27	3 2 1 4 4 3 1 3 4 0 1 2 2 4 3 4 4 2 3 3 4 4 4 1 3 3 3 1
14	21	19	0	21	1 2 1 2 0 3 3 2 1 1 3 3 3 1 2 1 4 3 2 2 0 3
15	22	19	0	21	3 0 3 2 4 4 2 3 3 0 3 1 3 3 3 0 4 3 2 2 0 1

$$\sigma_3(y) = \sum_{j=0}^{15} \frac{N_j(t)}{t^{a_j}(t+4)^{b_j}(t^2+4t+1)^{c_j}}\, y^j$$

j	a_j	b_j	c_j	$\deg N_j$	coefficients of N_j, from $t^{\deg N_j}$ down to t^0
0	4	12	0	12	1 3 2 0 4 3 3 2 4 1 3 1 4
1	5	13	0	16	1 2 3 3 3 0 3 0 0 4 1 3 2 4 1 4
2	7	13	0	17	1 1 0 3 3 0 2 2 1 0 0 0 2 2 3 4 2 1
3	9	12	0	17	1 3 2 0 1 0 1 3 4 0 0 4 0 4 0 1 3 4
4	11	14	0	19	4 3 0 4 4 2 1 4 0 1 2 4 0 4 4 1 3 4 0 4
5	14	17	0	23	3 0 1 4 0 2 3 4 1 1 3 3 4 3 3 1 4 0 2 4 4 0 2 4
6	15	18	0	25	4 1 2 1 4 3 0 1 1 3 2 0 4 4 1 2 0 0 3 3 1 1 0 2 0 3
7	16	18	0	23	4 4 2 2 3 1 0 0 4 1 0 3 0 1 2 3 1 4 2 2 3 0 4 1
8	16	18	0	23	1 3 3 0 3 1 0 1 0 1 3 1 2 1 0 1 4 4 4 0 2 3 2 4
9	13	18	0	18	2 0 0 1 4 2 3 3 3 3 2 3 2 3 4 3 3 4 2
10	14	18	0	18	2 3 3 3 2 3 1 0 2 1 2 3 4 3 3 3 1 3 4
11	15	18	0	18	4 0 0 0 2 3 4 1 1 1 2 3 4 2 4 2 4 0 2
12	16	18	0	16	2 1 4 3 0 1 3 4 2 0 3 3 4 3 0 1 4
13	16	18	0	16	2 3 1 4 3 0 4 2 0 4 2 1 0 4 2 3 1
14	13	18	0	11	3 4 0 4 0 2 1 4 3 3 0 4
15	13	18	0	10	4 3 1 4 2 2 0 1 1 0 1

The automorphism σ_4.
Orbits of $\langle\sigma_4\rangle$ on $X(\mathbb{F}_5)$: $\{P_1, P_2, P_3, P_4, P_5, P_6, P_7, P_8, P_9, P_{10}, P_{11}, P_{12}\}$.

$$\sigma_4(t) = \sum_{j=0}^{15} \frac{N_j(t)}{t^{a_j}(t+4)^{b_j}(t^2+4t+1)^{c_j}}\, y^j$$

j	a_j	b_j	c_j	$\deg N_j$	coefficients of N_j, from $t^{\deg N_j}$ down to t^0
0	17	10	0	25	41103033321124414123443424
1	18	11	0	28	40122142430104400314322044432
2	19	11	0	26	41043122200442331344423324 3
3	20	10	0	27	14002144024424100044413444 02
4	21	12	0	28	31433144324342342013020111034
5	27	15	0	37	41040330344214410220444400433413143424
6	28	16	0	37	14322223013210220431241222034224144432
7	29	16	0	35	31231434013311224043240443314222333 3
8	30	16	0	36	31232331421112131112130002123431020 24
9	26	16	0	31	33423213203142200311013344443304
10	27	16	0	31	43401204210400004020340420404412
11	28	16	0	30	14034244313044314234441344130 03
12	29	16	0	30	22203110432310401113330033124 22
13	30	16	0	29	21330122222433044203311431403 1
14	26	16	0	24	40220424441234040041423 03
15	27	16	0	22	41121130314413212412401

$$\sigma_4(y) = \sum_{j=0}^{15} \frac{N_j(t)}{t^{a_j}(t+4)^{b_j}(t^2+4t+1)^{c_j}}\, y^j$$

j	a_j	b_j	c_j	$\deg N_j$	coefficients of N_j, from $t^{\deg N_j}$ down to t^0
0	19	8	0	25	11424441021140220420202441
1	20	9	0	27	21421344432430233404023040 3 03
2	21	9	0	28	11020004211330002013420433432
3	22	10	0	28	20140021331134410242010000323
4	23	11	0	30	33001242120324042310321312241 34
5	29	13	0	36	32012103223140223214024411334330024 41
6	30	14	0	38	34042001401442121410123101301313420 4303
7	31	14	0	37	21242300340024331034410242334033132442
8	32	14	0	36	20233420442110412313430432412020023 41
9	28	14	0	31	22120313102410414123100421133211
10	29	14	0	31	13312024144341041422321002324023
11	30	14	0	31	34310444340314102110212322210022
12	31	14	0	30	22301011320132212311033423414 13
13	32	14	0	30	40242230311322231013302430322 14
14	28	14	0	24	41322132223301242040432 22
15	29	14	0	24	30231330114442322003201 44

The automorphism σ_5.
Orbits of $\langle\sigma_5\rangle$ on $X(\mathbb{F}_5)$: $\{P_1, P_5, P_7\},\ \{P_2, P_4, P_8\},\ \{P_3, P_6, P_{10}\},\ \{P_9, P_{11}, P_{12}\}$.

$$\sigma_5(t) = \sum_{j=0}^{15} \frac{N_j(t)}{t^{a_j}(t+4)^{b_j}(t^2+4t+1)^{c_j}}\, y^j$$

j	a_j	b_j	c_j	$\deg N_j$	coefficients of N_j, from $t^{\deg N_j}$ down to t^0
0	11	8	0	16	34022341033341201
1	12	9	0	20	1000421410014044004 33
2	13	9	0	20	44343141401332123444 2
3	14	9	0	19	201114210242304314 11
4	16	10	0	22	11321011430023130442411
5	21	13	0	29	31111024412044214232144232020 1
6	22	14	0	30	20330113414420113441034432110 33
7	23	14	0	29	30202331402242230331140230311 2
8	24	14	0	27	40400430433040324233002041 01
9	20	14	0	23	30013111342214101043222 1
10	21	14	0	23	23042420144221132042140 3
11	22	14	0	23	22234202314232432012404 2
12	23	14	0	22	23423130012343441423243
13	24	14	0	22	14243340302304344220404
14	20	14	0	16	41033343311004242
15	21	14	0	16	22113401112044134

$$\sigma_5(y) = \sum_{j=0}^{15} \frac{N_j(t)}{t^{a_j}(t+4)^{b_j}(t^2+4t+1)^{c_j}}\, y^j$$

j	a_j	b_j	c_j	$\deg N_j$	coefficients of N_j, from $t^{\deg N_j}$ down to t^0
0	15	6	0	21	1121412423421304134024
1	16	8	0	22	1100304230303040221 2143
2	17	5	0	19	434303240214201431 03
3	18	9	0	24	113441203343200332110 0313
4	19	10	0	25	33234311201211011242223044
5	25	11	0	31	10143344140444213112401422434024
6	26	13	0	33	14124203342422422100222104012221 43
7	27	13	0	32	43432322301343441111420231233 4302
8	28	13	0	31	23300234344202411441240424103121
9	24	13	0	26	31212002330410033140321 3141
10	25	13	0	26	32233423141412313202032 0413
11	26	13	0	26	10314144120213031224010 3332
12	27	13	0	25	21112220214242212203413003
13	28	13	0	25	31412021142431021021201434
14	24	13	0	19	4443121113140024 2032
15	25	13	0	19	1203433221411341 4214

The automorphism σ_6.

Orbits of $\langle\sigma_6\rangle$ on $X(\mathbb{F}_5)$: $\{P_1, P_6\},\ \{P_2, P_{11}\},\ \{P_3, P_7\},\ \{P_4, P_{12}\},\ \{P_5, P_{10}\},\ \{P_8, P_9\}$.

$$\sigma_6(t) = \sum_{j=0}^{15} \frac{N_j(t)}{t^{a_j}(t+4)^{b_j}(t^2+4t+1)^{c_j}}\, y^j$$

j	a_j	b_j	c_j	$\deg N_j$	coefficients of N_j, from $t^{\deg N_j}$ down to t^0
0	17	13	0	30	12313442334322234314411340120 31
1	18	14	0	30	14023112231041403203010332233 23
2	19	14	0	31	44210220243100102124032131112112
3	20	14	0	30	44200013343141434320012401432 32
4	21	15	0	32	11042330034342423134234042112 0421
5	27	18	0	39	42211021122104442243404334213422 14312031
6	28	19	0	41	40141014310010333022330300432411304112 3323
7	29	19	0	40	12002112244442211142322132332321032011022
8	30	19	0	38	33114024444202323014004333101340141 3431
9	26	19	0	34	31042123001221430400320043314210101
10	27	19	0	34	30200443314443123423403142320221343
11	28	19	0	34	41120022300331144401331434021333302
12	29	19	0	33	10220342131340102123020210024343 33
13	30	19	0	33	21223413341030304023110323403311 24
14	26	19	0	27	24104142434244214204411340 02
15	27	19	0	27	43400221331304301020314102 04

$$\sigma_6(y) = \sum_{j=0}^{15} \frac{N_j(t)}{t^{a_j}(t+4)^{b_j}(t^2+4t+1)^{c_j}}\, y^j$$

j	a_j	b_j	c_j	$\deg N_j$	coefficients of N_j, from $t^{\deg N_j}$ down to t^0
0	19	12	0	28	2230140334133321201144411 3311
1	20	13	0	31	1342423234304120412424012132211 3
2	21	13	0	32	1402121200041111403020330441422222
3	22	13	0	31	2002433041404032220420422442244 42
4	23	14	0	33	3040324402121210342321200134134401
5	29	17	0	40	4202223142330123212324131212112314341331 1
6	30	18	0	42	3341330231114003014310332321313421231222113
7	31	18	0	41	2232101443404440124040041311400212304124 32
8	32	18	0	40	2001021420431020320302013040013401000121 1
9	28	18	0	35	2032434020130020101313033042223220 31
10	29	18	0	35	1234134342043120321141441402034432 33
11	30	18	0	35	3320132304024442300014210131324301 12
12	31	18	0	34	2112002331012133334002140201340242 3
13	32	18	0	34	4203421420133421323223011103401334 4
14	28	18	0	28	4420110022401020200242444231 2
15	29	18	0	28	3401404203323142400041012232 4

The automorphism σ_7.
Orbits of $\langle\sigma_7\rangle$ on $X(\mathbb{F}_5)$: $\{P_1, P_5, P_7\},\ \{P_2, P_4, P_8\},\ \{P_3, P_6, P_{10}\},\ \{P_9, P_{11}, P_{12}\}$.

$$\sigma_7(t) = \sum_{j=0}^{15} \frac{N_j(t)}{t^{a_j}(t+4)^{b_j}(t^2+4t+1)^{c_j}}\, y^j$$

j	a_j	b_j	c_j	$\deg N_j$	coefficients of N_j, from $t^{\deg N_j}$ down to t^0
0	19	10	0	28	2413140041410422140210113101 1
1	20	11	0	29	1102414013102202223112411022 13
2	21	11	0	30	4220042044403103301340223403122
3	22	10	0	28	4111412223124101101230423214 3
4	23	12	0	31	4402401203030111103031311041010 1
5	29	15	0	39	1213323140304404022001312130333410431011
6	30	16	0	40	4434102443111101104244301302021423100221 3
7	31	16	0	39	1322420300044410334212232322220024420332
8	32	16	0	38	4224113230003400240124343232330330044 11
9	28	16	0	33	3204122232132120114134131222304231
10	29	16	0	33	4404300203232101011403043340012333
11	30	16	0	33	4431401310442231101232401041214012
12	31	16	0	32	4420412233044042203033213200040 23
13	32	16	0	32	2232033421212224121320242223101 44
14	28	16	0	26	3122112042322130311123112 12
15	29	16	0	26	4300433244131021040122001 24

$$\sigma_7(y) = \sum_{j=0}^{15} \frac{N_j(t)}{t^{a_j}(t+4)^{b_j}(t^2+4t+1)^{c_j}}\, y^j$$

j	a_j	b_j	c_j	$\deg N_j$	coefficients of N_j, from $t^{\deg N_j}$ down to t^0
0	18	10	0	28	3210312244410003042242423232 1
1	19	11	0	29	4224424232044112200222341034 43
2	20	11	0	29	4240041142113244042442331312 42
3	21	10	0	28	3133330011322131401013213312 3
4	22	12	0	31	4412134122201211022312440203031 1
5	28	15	0	39	1300433312032321444102111211244331032321
6	29	16	0	40	3014403444040041143223401142240320300344 3
7	30	16	0	39	2332324201432114202144401132332202002002
8	31	16	0	38	2201014333431431042404324000003210142 21
9	27	16	0	31	1324111242131440333423241000124 1
10	28	16	0	33	2340414230131221404312213021013213
11	29	16	0	33	3324431124111300022434240414344032
12	30	16	0	31	3121212242230442001234012112030 3
13	31	16	0	32	4020101344400201242023440302403 34
14	27	16	0	25	4034030010004240323410323 2
15	28	16	0	26	3220422213440423041400011 14

The automorphism σ_8.

Orbits of $\langle\sigma_8\rangle$ on $X(\mathbb{F}_5)$: $\{P_1, P_2, P_3, P_4, P_5, P_6, P_7, P_8, P_9, P_{10}, P_{11}, P_{12}\}$.

$$\sigma_8(t) = \sum_{j=0}^{15} \frac{N_j(t)}{t^{a_j}(t+4)^{b_j}(t^2+4t+1)^{c_j}}\, y^j$$

j	a_j	b_j	c_j	$\deg N_j$	coefficients of N_j, from $t^{\deg N_j}$ down to t^0
0	10	10	0	19	3331210114240132310 1
1	11	11	0	20	4123432442424334222 33
2	12	11	0	21	4233241433230440041302
3	13	10	0	19	2333013440412434431 3
4	14	12	0	22	4140004000414322133034 1
5	20	15	0	30	4122001201021122402114122123101
6	21	16	0	28	3033414301411422304233032223 3
7	22	16	0	28	2030004301232043344213233341 2
8	23	16	0	28	4320401200424020230000314200 1
9	19	16	0	24	2031300034430312200403121
10	20	16	0	24	4300120130403144122200103
11	21	16	0	23	2310022023020100130213 42
12	22	16	0	23	2120131004331414422134 43
13	23	16	0	22	3023210014302141143200 4
14	19	16	0	17	4140000230030410 42
15	20	16	0	16	4332220114442323 4

$$\sigma_8(y) = \sum_{j=0}^{15} \frac{N_j(t)}{t^{a_j}(t+4)^{b_j}(t^2+4t+1)^{c_j}}\, y^j$$

j	a_j	b_j	c_j	$\deg N_j$	coefficients of N_j, from $t^{\deg N_j}$ down to t^0
0	12	10	0	22	1142314041410124333433 4
1	13	11	0	22	1103033344130041410341 2
2	14	11	0	21	3141133202234111001013
3	15	10	0	22	1132441020340144141123 2
4	16	12	0	23	1331120432023340002403 44
5	22	15	0	32	2424004302340202003031242110343 34
6	23	16	0	33	4101100031424140223344400330203412
7	24	16	0	32	1310041411001110020010013244012 03
8	25	16	0	30	4102014332220031244312032342434
9	21	16	0	26	1230411304333124233320324 14
10	22	16	0	25	1041223332222320021040014 2
11	23	16	0	26	4434242420431143230223322 23
12	24	16	0	25	2112011323241024403341200 2
13	25	16	0	25	2023041343041014404341412 1
14	21	16	0	19	4401232404114424302 3
15	22	16	0	19	4134440213220140134 1

The automorphism σ_9.
Orbits of $\langle\sigma_9\rangle$ on $X(\mathbb{F}_5)$: $\{P_1, P_2, P_3, P_4, P_5, P_6, P_7, P_8, P_9, P_{10}, P_{11}, P_{12}\}$.

$$\sigma_9(t) = \sum_{j=0}^{15} \frac{N_j(t)}{t^{a_j}(t+4)^{b_j}(t^2+4t+1)^{c_j}}\, y^j$$

j	a_j	b_j	c_j	$\deg N_j$	coefficients of N_j, from $t^{\deg N_j}$ down to t^0
0	11	9	0	19	20223203003323220041
1	12	10	0	21	10232411403134331130 22
2	14	10	0	22	43044012401404024314224
3	16	9	0	22	10210210224412030002311
4	18	11	0	24	12024440030100202122344 41
5	21	14	0	30	44140341012313422240134120 00431
6	22	15	0	31	21402204441333021420233442140423
7	23	15	0	30	34011222243302132043022424341 22
8	24	15	0	29	21030321332200224042401113 1031
9	20	15	0	24	44304344303024411033214 01
10	21	15	0	24	34031401144244000001204 43
11	22	15	0	24	20214442214014402242432 02
12	23	15	0	22	22133304432333142141233
13	24	15	0	23	12224130440004404401302 4
14	20	15	0	15	43334020013403 02
15	21	15	0	17	241414002441400104

$$\sigma_9(y) = \sum_{j=0}^{15} \frac{N_j(t)}{t^{a_j}(t+4)^{b_j}(t^2+4t+1)^{c_j}}\, y^j$$

j	a_j	b_j	c_j	$\deg N_j$	coefficients of N_j, from $t^{\deg N_j}$ down to t^0
0	12	10	0	21	41433400111102210321 11
1	13	11	0	23	12322323404242141133141 3
2	14	11	0	23	32420320023121110041341 2
3	15	9	0	21	31321001240002134133 21
4	17	12	0	25	24030142030232234140044141
5	22	15	0	32	22324330032143131344344202333111 1
6	23	16	0	33	40401330344002023004423240034300 13
7	24	16	0	32	12313112432332414400341440132 1032
8	25	16	0	31	11203020223312114340242323114011
9	21	16	0	26	32231020010421010124014133 1
10	22	16	0	25	30112113312221323130432133
11	23	16	0	26	44332040244214131140333321 2
12	24	16	0	25	20413313231441124030233323
13	25	16	0	25	24412222434134430124100044
14	21	16	0	19	41002312343314140412
15	22	16	0	19	40331222143032413024

The automorphism σ_{10}.
Orbits of $\langle\sigma_{10}\rangle$ on $X(\mathbb{F}_5)$: $\{P_1, P_3, P_5, P_6, P_7, P_{10}\}$, $\{P_2, P_4, P_8, P_9, P_{11}, P_{12}\}$.

$$\sigma_{10}(t) = \sum_{j=0}^{15} \frac{N_j(t)}{t^{a_j}(t+4)^{b_j}(t^2+4t+1)^{c_j}}\, y^j$$

j	a_j	b_j	c_j	$\deg N_j$	coefficients of N_j, from $t^{\deg N_j}$ down to t^0
0	10	10	0	20	1 0 4 2 3 2 4 0 2 2 0 3 3 1 4 4 4 3 2 4 1
1	11	11	0	19	2 0 1 1 2 1 0 2 4 3 4 3 3 2 3 2 1 2 0 3
2	12	11	0	19	2 1 4 0 1 3 3 4 1 4 2 0 2 0 3 3 0 0 3 2
3	13	10	0	20	1 2 0 3 4 2 1 4 3 3 3 4 0 4 2 4 4 2 0 0 3 3
4	14	12	0	21	2 1 1 2 2 3 0 1 3 0 2 1 1 4 2 0 3 1 4 4 3 1
5	20	15	0	29	2 2 4 2 3 0 1 1 0 4 2 3 3 3 0 0 1 2 4 3 3 4 2 4 2 4 3 2 4 1
6	21	16	0	31	1 2 2 2 1 4 3 0 1 0 2 4 2 2 1 1 3 3 4 2 1 4 0 3 1 3 4 2 1 2 0 3
7	22	16	0	30	4 0 1 0 3 4 0 4 3 3 3 1 0 1 0 1 2 3 0 2 4 2 2 2 3 4 3 2 0 4 2
8	23	16	0	29	3 3 4 4 3 4 2 4 1 1 0 3 1 2 2 4 2 1 3 2 4 2 1 2 3 1 3 1 4 1
9	19	16	0	24	3 2 0 4 0 4 4 3 3 4 2 2 3 1 0 2 1 1 0 2 0 0 0 1 1
10	20	16	0	24	2 3 3 2 3 2 4 4 2 0 2 4 0 3 4 1 1 1 3 1 3 2 4 2 3
11	21	16	0	24	1 4 0 4 3 3 2 0 0 2 3 4 3 0 1 1 2 4 3 1 1 4 1 2 2
12	22	16	0	23	4 3 4 1 0 1 0 1 2 1 1 0 1 2 0 4 2 3 1 3 1 3 1 3
13	23	16	0	23	3 1 2 0 4 0 3 4 3 0 3 4 3 2 4 0 4 3 4 2 1 4 1 4
14	19	16	0	17	3 4 0 4 1 0 0 3 4 3 3 2 1 0 2 3 2 2
15	20	16	0	17	1 4 2 2 0 0 0 3 4 3 2 1 0 0 3 3 4 4

$$\sigma_{10}(y) = \sum_{j=0}^{15} \frac{N_j(t)}{t^{a_j}(t+4)^{b_j}(t^2+4t+1)^{c_j}}\, y^j$$

j	a_j	b_j	c_j	$\deg N_j$	coefficients of N_j, from $t^{\deg N_j}$ down to t^0
0	12	11	0	21	1 0 3 0 4 4 3 4 1 0 4 1 2 4 0 1 1 0 2 1 1 1
1	13	12	0	22	3 0 2 1 0 0 4 4 4 1 2 4 0 3 2 0 2 1 3 3 0 1 3
2	14	12	0	23	4 4 0 1 4 3 3 0 2 1 0 3 4 1 4 0 0 0 0 0 0 3 2 2
3	15	11	0	23	4 3 0 1 3 4 2 1 0 1 4 0 1 2 2 2 1 2 2 0 1 4 4 3
4	16	13	0	24	2 2 4 1 1 3 2 0 0 0 4 4 0 2 1 1 1 2 4 2 3 0 2 0 1
5	22	16	0	32	2 2 2 2 4 4 1 1 0 3 0 0 3 0 0 1 0 4 3 1 4 2 3 1 2 4 0 4 0 2 1 1 1
6	23	17	0	34	4 4 4 2 3 3 1 1 1 1 3 3 4 0 4 4 3 3 3 2 1 4 1 4 3 0 3 2 2 0 3 3 0 1 3
7	24	17	0	33	1 4 3 0 0 3 1 0 2 1 0 0 2 2 4 2 3 3 0 2 0 1 3 3 1 0 1 3 4 4 3 0 3 2
8	25	17	0	32	2 4 2 0 1 0 3 2 4 2 4 3 0 2 0 0 2 3 4 4 3 3 3 0 1 1 3 3 1 0 0 1 1
9	21	17	0	27	2 3 4 3 4 3 2 1 0 3 0 4 2 1 0 3 4 1 2 4 0 3 4 4 2 3 3 1
10	22	17	0	27	3 3 3 0 4 0 0 2 4 1 4 1 0 3 2 1 3 1 4 0 1 2 4 0 0 1 3 3
11	23	17	0	27	4 2 0 3 3 2 1 0 0 1 3 2 3 0 2 1 3 0 4 0 2 4 2 0 0 2 1 2
12	24	17	0	26	1 3 0 2 1 2 0 1 0 2 2 0 1 0 0 1 3 3 3 1 4 4 1 1 3 2 3
13	25	17	0	26	2 2 4 0 2 3 4 1 3 1 0 2 1 2 2 1 4 4 1 0 3 4 0 4 0 4 4
14	21	17	0	20	2 3 2 4 2 3 2 3 3 2 0 0 4 4 4 0 2 2 4 1 2
15	22	17	0	20	4 2 0 4 4 2 1 3 4 0 0 3 3 3 3 4 4 2 0 2 4

The automorphism σ_{11}.

Orbits of $\langle\sigma_{11}\rangle$ on $X(\mathbb{F}_5)$: $\{P_1, P_2, P_6, P_{11}\},\ \{P_3, P_4, P_7, P_{12}\},\ \{P_5, P_8, P_9, P_{10}\}$.

$$\sigma_{11}(t) = \sum_{j=0}^{15} \frac{N_j(t)}{t^{a_j}(t+4)^{b_j}(t^2+4t+1)^{c_j}}\, y^j$$

j	a_j	b_j	c_j	$\deg N_j$	coefficients of N_j, from $t^{\deg N_j}$ down to t^0
0	12	12	2	26	2 1 1 3 2 4 1 1 1 4 4 1 1 4 4 0 4 4 4 0 1 4 1 4 1 3 4
1	13	13	2	29	4 2 3 0 4 1 4 1 1 1 4 0 3 0 0 1 2 4 0 1 0 1 3 2 3 1 2 3 1 2
2	14	13	2	29	4 2 2 0 2 3 2 3 3 4 0 0 2 0 3 3 1 3 3 1 1 3 4 2 2 4 1 1 1 3
3	15	13	2	28	4 2 1 2 4 2 3 2 1 3 4 4 3 3 1 1 4 4 1 3 1 4 3 3 4 2 2 4 3
4	16	14	2	30	1 2 2 2 2 0 3 0 0 0 0 1 0 4 2 3 4 1 3 3 2 4 3 0 0 2 3 2 1 4 4
5	22	17	2	38	2 4 4 2 1 4 2 4 3 0 1 2 0 4 2 4 3 4 2 2 2 4 0 2 4 2 4 2 3 0 2 1 4 1 1 4 1 3 4
6	23	18	2	39	2 2 0 1 2 1 0 3 3 1 1 3 4 3 2 2 4 0 1 4 0 3 4 4 0 0 3 3 0 1 0 4 2 3 4 1 2 3 1 2
7	24	18	2	38	3 1 4 1 3 3 1 1 2 4 1 1 1 1 3 2 1 3 3 1 4 3 0 0 3 0 0 1 0 3 0 1 4 2 2 4 3 0 3
8	25	18	2	37	3 4 0 3 0 3 3 3 4 2 3 3 4 1 3 0 4 1 0 4 3 1 4 0 1 4 1 3 4 4 1 1 1 4 2 2 3 4
9	21	18	2	32	1 1 3 3 2 4 4 1 1 2 1 4 1 0 4 1 1 1 1 0 2 2 3 3 1 1 3 1 1 3 2 1 4
10	22	18	2	32	1 0 4 2 3 2 3 0 3 3 4 1 2 3 3 1 3 0 2 0 1 1 0 0 3 4 1 0 3 0 0 4 2
11	23	18	2	32	2 0 4 1 3 0 4 1 2 4 4 2 2 3 2 4 4 4 3 2 3 4 3 1 4 1 0 0 2 4 3 2 3
12	24	18	2	31	4 2 1 1 3 3 2 4 4 4 1 0 1 3 4 1 0 2 2 4 3 0 3 3 0 0 4 3 4 4 0 2
13	25	18	2	31	1 2 1 0 2 3 2 1 0 4 0 2 2 4 3 1 3 4 3 2 3 2 1 1 0 3 1 4 3 2 1
14	21	18	2	25	3 4 1 4 2 4 4 3 2 2 0 2 4 4 1 4 0 4 2 3 4 4 0 1 2 3
15	22	18	2	25	2 4 1 0 3 2 0 1 0 1 3 4 4 2 1 3 2 3 2 1 1 1 0 0 4 1

$$\sigma_{11}(y) = \sum_{j=0}^{15} \frac{N_j(t)}{t^{a_j}(t+4)^{b_j}(t^2+4t+1)^{c_j}}\, y^j$$

j	a_j	b_j	c_j	$\deg N_j$	coefficients of N_j, from $t^{\deg N_j}$ down to t^0
0	11	13	3	30	1 2 0 0 3 3 3 2 0 0 3 0 2 1 0 4 4 4 2 1 3 4 4 4 4 2 1 3 4 1 4
1	12	14	3	30	2 0 0 3 3 1 4 4 1 2 2 1 4 2 3 1 4 0 3 4 0 1 0 0 1 4 1 0 1 0 2
2	13	14	3	30	4 4 4 2 3 4 0 2 4 3 0 4 2 4 0 2 1 4 2 4 0 3 4 0 4 3 1 3 2 2 3
3	14	14	3	31	4 1 4 3 1 4 0 1 3 4 3 4 4 0 1 3 3 3 2 4 2 4 3 0 2 1 3 0 3 0 0 3
4	15	15	3	32	3 1 4 4 2 3 2 1 3 3 4 4 1 0 0 1 3 0 1 2 1 2 0 3 4 3 4 2 3 1 3 1 4
5	21	18	3	40	4 0 0 1 2 2 3 2 2 2 4 4 3 4 3 1 2 0 2 2 0 2 1 1 0 2 1 2 1 1 3 0 4 4 1 0 1 3 4 1 4
6	22	19	3	41	1 0 3 3 2 1 2 4 4 1 1 3 1 2 2 1 4 0 0 0 2 4 1 1 4 3 4 2 4 3 1 1 2 2 4 2 1 0 0 1 0 2
7	23	19	3	40	4 0 4 3 3 3 1 0 0 4 1 3 2 0 3 3 1 3 2 0 3 4 0 3 0 1 2 2 1 4 2 4 4 2 1 3 0 1 2 1 3
8	24	19	3	38	1 2 1 3 0 0 2 3 2 3 3 4 3 3 4 4 4 1 3 4 1 4 1 1 3 3 3 2 1 0 1 0 3 2 3 3 0 1 4
9	20	19	3	34	4 3 2 0 0 0 0 3 2 2 3 4 1 4 1 2 4 4 1 4 3 4 3 0 3 1 2 3 0 0 3 3 1 4 4
10	21	19	3	34	1 3 1 1 4 2 0 0 3 2 2 4 0 4 0 3 0 0 4 2 3 0 3 2 1 2 2 3 2 0 0 1 4 3 2
11	22	19	3	34	1 1 2 3 4 1 0 4 1 4 4 3 2 0 3 1 1 1 4 3 2 3 3 2 4 2 3 0 1 0 4 2 1 3 3
12	23	19	3	33	1 1 0 4 4 3 3 0 1 4 3 4 3 1 4 3 2 0 0 4 4 4 2 4 4 2 2 0 1 2 1 0 4 2
13	24	19	3	33	3 2 4 0 4 1 0 0 0 2 2 4 0 1 3 3 2 0 3 0 2 3 3 1 1 2 3 3 0 4 3 0 4 1
14	20	19	3	27	2 2 1 1 1 4 1 0 4 0 2 2 3 3 1 3 4 3 1 2 2 2 2 4 4 3 3
15	21	19	3	27	1 1 0 4 2 2 2 1 3 2 0 2 0 4 0 2 2 3 2 3 1 2 4 1 4 1 1 1

The automorphism σ_{12}.
Orbits of $\langle\sigma_{12}\rangle$ on $X(\mathbb{F}_5)$: $\{P_1, P_2, P_3, P_4, P_5, P_6, P_7, P_8, P_9, P_{10}, P_{11}, P_{12}\}$.

$$\sigma_{12}(t) = \sum_{j=0}^{15} \frac{N_j(t)}{t^{a_j}(t+4)^{b_j}(t^2+4t+1)^{c_j}}\, y^j$$

j	a_j	b_j	c_j	$\deg N_j$	coefficients of N_j, from $t^{\deg N_j}$ down to t^0
0	11	12	0	23	1 4 0 4 4 4 2 2 2 1 1 0 1 3 0 1 3 2 1 4 1 1 3 4
1	12	13	0	24	4 2 1 0 0 3 0 4 2 4 3 1 2 4 4 4 2 0 4 3 1 1 3 3 1 2
2	13	13	0	24	1 2 0 0 3 2 3 0 4 0 0 2 0 0 4 4 1 2 2 3 0 0 1 1 3
3	14	12	0	23	4 2 2 3 1 0 1 1 1 3 0 2 4 2 1 0 1 0 1 1 0 1 3 2
4	15	14	0	25	2 3 3 4 2 4 0 0 4 4 4 4 3 0 0 3 1 1 4 0 3 4 4 1 4 4
5	21	17	0	33	2 2 3 3 4 0 2 4 0 0 2 2 1 0 0 1 2 3 0 4 2 4 2 2 4 0 4 0 3 4 1 1 3 4
6	22	18	0	34	4 3 3 4 1 0 2 3 3 4 0 3 4 0 4 3 2 4 3 4 3 2 3 1 3 0 4 3 4 2 1 3 3 1 2
7	23	18	0	33	1 4 3 3 3 3 3 4 3 2 3 0 2 3 1 2 4 4 4 3 3 3 1 3 4 2 4 3 1 3 0 3 3 0 3
8	24	18	0	32	2 1 2 3 1 4 4 3 4 4 2 3 3 4 4 3 1 4 3 0 3 3 3 3 1 0 2 1 2 4 2 3 4
9	20	18	0	27	3 4 1 1 0 0 3 1 2 1 3 1 4 1 0 2 2 0 2 1 3 4 4 3 0 2 1 4
10	21	18	0	26	2 2 0 0 0 3 2 0 1 0 2 3 1 1 4 1 0 0 2 4 0 2 2 1 0 4 2
11	22	18	0	27	4 2 2 0 3 4 2 0 3 4 4 1 2 4 1 4 1 0 2 2 1 0 3 1 3 3 2 3
12	23	18	0	26	2 3 1 1 2 1 4 2 2 1 4 0 2 2 3 0 3 4 2 4 4 4 0 0 4 0 2
13	24	18	0	26	2 3 0 4 4 0 4 3 4 3 1 1 0 1 2 3 1 4 0 3 2 3 3 2 3 2 1
14	20	18	0	20	4 0 1 2 1 1 3 4 2 3 3 2 2 1 4 1 3 4 1 2 3
15	21	18	0	20	4 3 1 3 4 4 4 2 3 3 3 3 1 3 1 1 4 4 3 0 4 1

$$\sigma_{12}(y) = \sum_{j=0}^{15} \frac{N_j(t)}{t^{a_j}(t+4)^{b_j}(t^2+4t+1)^{c_j}}\, y^j$$

j	a_j	b_j	c_j	$\deg N_j$	coefficients of N_j, from $t^{\deg N_j}$ down to t^0
0	12	13	0	25	1433213333022114231401 4014
1	13	14	0	26	4301420101312120401020214 02
2	14	14	0	26	1341211212232434434301344 23
3	15	13	0	25	4401013222301344222132312 2
4	16	15	0	27	1403244134412034414121224 224
5	22	18	0	35	2020133202042140212443330030212 14014
6	23	19	0	36	4412002434213300433011040444230121 402
7	24	19	0	35	1022334003330023123303300333310344 13
8	25	19	0	33	2312142223233021034033321012224 114
9	21	19	0	29	2341001410214212303400030012 44
10	22	19	0	28	2130442014413444234004303423 2
11	23	19	0	29	4030043223031031131312111433 33
12	24	19	0	28	2024000203301244201103232334 2
13	25	19	0	28	2003103411404041134320040244 1
14	21	19	0	22	4040004341102143104013 3
15	22	19	0	22	4422131130332001132101 1